

# Low scaling BSE implementation in the exciting code

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## Summary

Solving the Bethe-Salpeter Equation (BSE) is essential for understanding excited-state systems, but often challenging to converge or even computationally prohibitive for large systems. We implement a matrix-free BSE solver leveraging Interpolative Separable Density Fitting (ISDF) to interpolate electron-hole interaction kernels together with the Lanczos algorithm for diagonalization, avoiding full matrix setup. The scaling of our implementation is bounded by  $\mathcal{O}(N_o N_u N_k \log N_k)$  and is thus a massive improvement over methods that set up the whole matrix, scaling with at least  $\mathcal{O}((N_o N_u N_k)^3)$ , where  $N_o$  and  $N_u$  are the numbers of occupied and unoccupied states, respectively, and  $N_k$  is the number  $k$ -points.

## Theoretical background

The Bethe-Salpeter Equation (BSE) within many body perturbation theory (MBPT) provides the state-of-the-art frame work for describing light-matter interaction. In particular, it is used to obtain optical absorption spectra, including the effects of excitons, which are bound electron-hole states. By expanding the electron-hole wavefunctions in the transition basis, solving the BSE can be reduced to a Schrödinger like equation. Setting up and diagonalizing the Bethe-Salpeter Hamiltonian (BSH) are the computationally expensive tasks (Vorwerk et al., 2019). The BSH is given as

$$H^{BSH} = D + \gamma V - W,$$

where  $\gamma = 2$  gives the spin-singlet and  $\gamma = 0$  spin-triplet channel, respectively. The diagonal term  $D$  is given by the differences of the one-particle energies of the occupied ( $o$ ) and unoccupied ( $u$ ) states:

$$D_{ouk,o'u'k'} = (\varepsilon_{uk} - \varepsilon_{ok}) \delta_{oo'} \delta_{uu'} \delta_{kk'}.$$

The matrix elements of the repulsive exchange interaction  $V$  and the attractive screened Coulomb interaction  $W$  are calculated by solving integrals of the form

$$V_{ouk,o'u'k'} = \int d^3r \int d^3r' \frac{u_{ok}(\mathbf{r}) u_{uk}^*(\mathbf{r}) u_{o'k'}^*(\mathbf{r}') u_{u'k'}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|},$$

$$W_{ouk,o'u'k'} = \int d^3r \int d^3r' u_{uk}^*(\mathbf{r}) u_{u'k'}(\mathbf{r}) W_{k-k'}(\mathbf{r}, \mathbf{r}') u_{ok}(\mathbf{r}') u_{o'k'}^*(\mathbf{r}'),$$

where  $W_{k-k'}(\mathbf{r}, \mathbf{r}')$  is the statically screened Coulomb potential.  $u_{ik}(\mathbf{r})$  is the periodic part of the one-particle wavefunction of state  $i$  at the reciprocal lattice point  $\mathbf{k}$ . In general, we cannot see, which of the matrix elements will be zero, thus we need to compute all of them. Therefore, setting up the full BSH scales with  $\mathcal{O}(N_o^2 N_u^2 N_k^2)$  and diagonalizing it directly with  $\mathcal{O}(N_o^3 N_u^3 N_k^3)$ .

33 The primary scaling bottleneck stems from evaluating the matrix elements of the interaction  
 34 kernels. To mitigate this, we reformulate the wavefunction products using Interpolative  
 35 Separable Density Fitting (ISDF) Lu & Ying (2016). Specifically, we approximate these  
 36 products on a discrete real-space grid,  $\{\mathbf{r}\}$ , by expressing them as superpositions of values  
 37 evaluated on a smaller interpolation grid,  $\{\mathbf{r}_\mu\} \subset \{\mathbf{r}\}$ :

$$u_{ik}^*(\mathbf{r})u_{jk'}(\mathbf{r}) \approx \sum_{\mu=1}^{N_\mu} \zeta_\mu(\mathbf{r})u_{ik}^*(\mathbf{r}_\mu)u_{jk'}(\mathbf{r}_\mu),$$

38 where  $N_\mu$  is the number of interpolation points, and  $\zeta_\mu(\mathbf{r})$  are the expansion coefficients.  
 39 Due to the tensor product structure of  $u_{ik}^*(\mathbf{r})u_{jk'}(\mathbf{r})$ ,  $\zeta_\mu(\mathbf{r})$  can be computed efficiently, and  
 40 the scaling is bounded by  $\mathcal{O}(N_\mu^3)$  (Hu et al., 2017). We observe that we can always choose  
 41  $N_\mu \ll N_o N_u N_k$ , thus ISDF is never a bottleneck. The interpolation points are computed  
 42 efficiently with centroidal Voronoi tessellation within  $\mathcal{O}(N_\mu N_r)$  (Dong et al., 2018).

43 Inserting ISDF in the equations for the interaction kernels yields

$$V_{ouk,o'u'k'} \approx \frac{1}{N_k^2} \sum_{\mu=1}^{N_\mu^V} \sum_{\nu=1}^{N_\mu^V} u_{ok}(\mathbf{r}_\mu^V)u_{uk}^*(\mathbf{r}_\mu^V)\tilde{V}_{\mu\nu}u_{o'k'}^*(\mathbf{r}_\nu^V)u_{u'k'}(\mathbf{r}_\nu^V),$$

$$W_{ouk,o'u'k'} \approx \frac{1}{N_k^2} \sum_{\mu=1}^{N_\mu^{W_u}} \sum_{\nu=1}^{N_\mu^{W_o}} u_{uk}^*(\mathbf{r}_\mu^{W_u})u_{u'k'}(\mathbf{r}_\mu^{W_u})\tilde{W}_{\mu\nu,k-k'}u_{ok}(\mathbf{r}_\nu^{W_o})u_{o'k'}^*(\mathbf{r}_\nu^{W_o}),$$

45 where we have shifted the integration from the wavefunction pairs to the interpolation  
 46 coefficients such that

$$\tilde{V}_{\mu\nu} = \int_{\Omega^l \times \Omega^l} d\mathbf{r}d\mathbf{r}' \zeta_\mu^* V(\mathbf{r}, \mathbf{r}') \zeta_\nu^V(\mathbf{r}'),$$

$$\tilde{W}_{\mu\nu,k-k'} = \int_{\Omega^l \times \Omega^l} d\mathbf{r}d\mathbf{r}' \zeta_\mu^* W_{k-k'}(\mathbf{r}, \mathbf{r}') \zeta_\nu^{W_o}(\mathbf{r}').$$

48 Note that there are three different wavefunction pairings, i.e.  $u_{ok}^*(\mathbf{r})u_{uk}(\mathbf{r})$  for the exchange  
 49 kernel (V) and  $u_{ok}^*(\mathbf{r})u_{o'k'}(\mathbf{r})$ ,  $u_{uk}^*(\mathbf{r})u_{u'k'}(\mathbf{r})$  for the screened kernel W. Each pairing  
 50 requires a separate ISDF calculation, denoted by the superscripts V,  $W_o$ , and  $W_u$ . Since  
 51 the number of combinations may vary, the number of interpolation points required may also  
 52 vary. Reformulating the matrix elements in this way alone does not improve scaling with  
 53 respect to the system size. To achieve this, we combine it with an iterative solver, here  
 54 with the Lanczos algorithm. This class of algorithms constructs an approximation to the  
 55 eigenvalues and eigenvectors by iteratively applying matrix-vector multiplications. Applying  
 56 the interaction kernels in their interpolated forms to a vector  $X$  of dimension  $N_o N_u N_k$  allows  
 57 efficient computation by rearranging the summations to exploit a separable structure of the  
 58 kernels. For the exchange kernel we get

$$[V \cdot X]_{ouk} = \frac{1}{N_k} \sum_{\mu=1}^{N_\mu^V} u_{uk}^*(\mathbf{r}_\mu^V)u_{ok}(\mathbf{r}_\mu^V) \left\{ \sum_{\nu=1}^{N_\mu^V} \tilde{V}_{\mu\nu} \left[ \sum_{k'} \left( \sum_{u'} u_{u'k'}(\mathbf{r}_\nu^V) \left[ \sum_{o'} u_{o'k'}^*(\mathbf{r}_\nu^V) \cdot X_{o'u'k'} \right] \right) \right] \right\},$$

59 where we first compute the sums over  $o'$ ,  $u'$ , and  $k'$  to get a term that depends only on  $\mathbf{r}_\nu^V$   
 60 with a complexity of  $\mathcal{O}(N_\mu^V(N_o N_u N_k + N_u N_k))$ . The remaining sums can be computed with  
 61  $\mathcal{O}((N_\mu^V)^2 N_\mu^V N_o N_u N_k)$ , so the complexity of computing  $V \cdot X$  is bounded by  $\mathcal{O}((N_\mu^V)^2 +$   
 62  $N_\mu^V N_o N_u N_k)$ . Applying the screened kernel to  $X$ , after reordering the sums, we get

$$[W \cdot X]_{ouk} = \frac{1}{N_k} \sum_{\nu=1}^{N_\mu^{W_o}} u_{ok}(\mathbf{r}_\nu^{W_o}) \left\{ \sum_{\mu=1}^{N_\mu^{W_u}} u_{uk}^*(\mathbf{r}_\mu^{W_u}) \sum_{k'} \left[ \tilde{W}_{\mu\nu,k-k'} \right. \right.$$

63

$$\times \left( \sum_{u'} u_{u'k'}(\mathbf{r}_{\mu}^{W_u}) \left[ \sum_{o'} u_{o'k'}^*(\mathbf{r}_{\nu}^{W_o}) X_{o'u'k'} \right] \right) \Bigg\}.$$

64 Here we exploit the separable structure of the decomposition so that the terms depending on  
65  $\mathbf{k}$  and  $\mathbf{k}'$  are on the left and right of  $\tilde{W}_{\mu\nu, \mathbf{k}-\mathbf{k}'}$ . The evaluation of the two innermost sums  
66 over  $o'$  and  $u'$  to  $A_{\mu\nu}^{k'}$  scales with  $\mathcal{O}(N_{\mu}^{W_u} N_o N_u N_k + N_{\mu}^{W_o} N_{\mu}^{W_u} N_u N_k)$ . Then the sum over  
67  $\mathbf{k}'$  reads as a discrete convolution

$$\sum_{\mathbf{k}'} W_{\mathbf{k}-\mathbf{k}'} A_{\mu\nu}^{k'},$$

68 which can be efficiently evaluated with fast Fourier transforms simultaneously within  
69 the  $\mathcal{O}(Nk \log Nk)$  scaling for each  $\mu\nu$  pair. The remaining summations scale with  
70  $\mathcal{O}(N_{\mu}^{W_o} N_{\mu}^{W_u} N_u N_k)$ . So the complexity for the computation of  $W \cdot X$  is bounded by  
71  $\mathcal{O}(N_{\mu}^{W_u} N_o N_u N_k + N_{\mu}^{W_o} N_{\mu}^{W_u} N_u N_k + N_{\mu}^{W_o} N_{\mu}^{W_u} N_k \log N_k)$ .

## 72 Statement of need

73 Due to the unfavorable scaling of solving the BSE directly, many interesting problems such as  
74 complex materials with large unit cells or systems requiring a dense Brillouin-zone sampling  
75 are not feasible. Even though Henneke and coworkers (Henneke et al., 2020) have already  
76 described the new algorithm and demonstrated the scaling improvement, an easy-to-use and  
77 scalable implementation was still missing. We have implemented and fully integrated this  
78 approach in the existing BSE infrastructure of the all-electron, full-potential package exciting  
79 (Gulans et al., 2014). Users can now easily choose which algorithm they prefer to use and  
80 have the full suite of exciton analysis implemented in exciting at hand.

## 81 Results

82 For computing the ISDF, two new parameters,  $n_r$  and  $c_{\mu}$ , are introduced.  $n_r$  is the real-space  
83 sampling density for  $u_{ik}(\mathbf{r})$  and is defined as

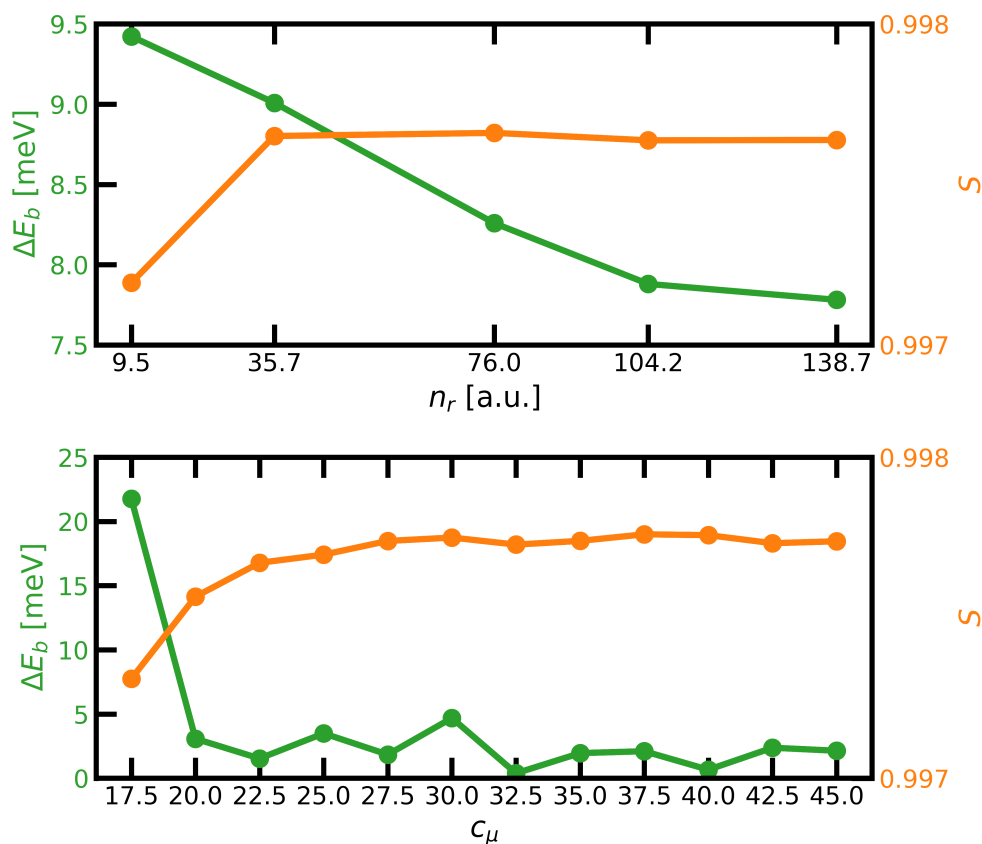
$$n_r = \frac{N_r}{\Omega},$$

84 where  $N_r$  is the number of  $\mathbf{r}$ -points and  $\Omega$  the unit cell volume. The sampling is chosen to be  
85 regular such that the distance between the sampling points in each lattice direction is as similar  
86 as possible. The dimensionless parameter  $c_{\mu}$  is used to control the number of interpolation  
87 points and is defined as

$$N_{\mu} = c_{\mu} \sqrt{\sqrt{N_{\text{pairs}}}},$$

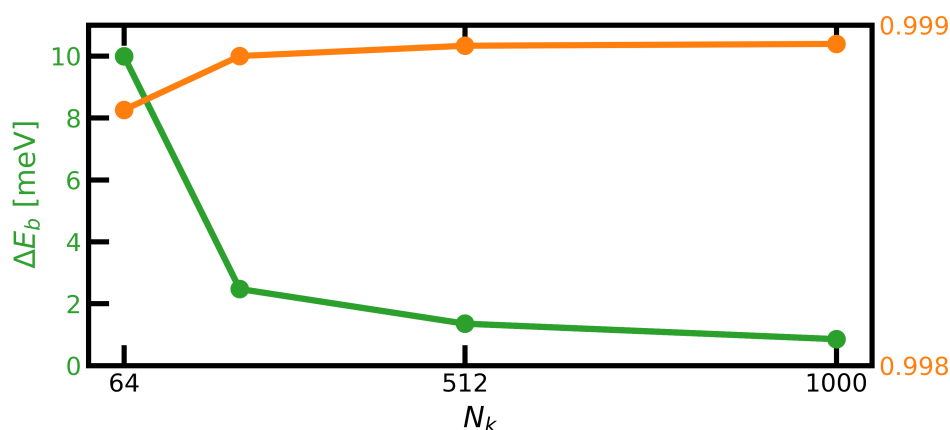
88 where  $N_{\text{pairs}}$  refers to the number of wave function pairs for which ISDF is computed.  
89 Note that  $N_{\text{pairs}}$  depends on  $N_k$ . The double square-root dependence ensures that the  
90 overall scaling remains below  $\mathcal{O}(N_k^2)$ . In Fig.(??) we present, for the example of diamond,  
91 the difference in exciton binding energies obtained with the new implementation and a  
92 reference calculation. The reference, based on the direct implementation, sets up and  
93 diagonalizes the full BSH and depends neither on  $n_r$  nor  $c_{\mu}$ . The results are shown  
94 as functions of  $n_r$  and  $c_{\mu}$  for a small  $\mathbf{k}$ -grid of  $2 \times 2 \times 2$ . Additionally, we show the  
95 spectral similarities compared to the reference calculation as functions of  $n_r$  and  $c_{\mu}$ . For  
96 both parameters, both properties converge as their values increase. To find the optimal  
97 interpolation grid for ISDF, we first converge  $n_r$ , then  $c_{\mu}$ . In our example,  $n_r = 138$   
98 [a.u.] and  $c_{\mu} = 40.0$  yield converged results. This corresponds to a real-space sampling

99 of  $22 \times 22 \times 22$  and numbers of interpolation points  $N_\mu^V = 202$ ,  $N_\mu^{W_o} = 322$ , and  $N_\mu^{W_u} = 360$ .



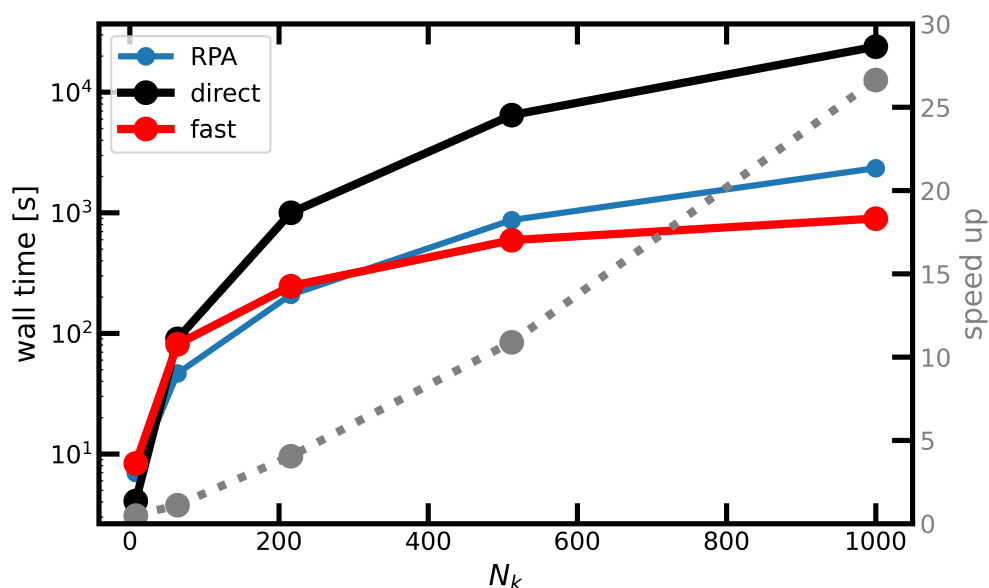
100

101 In Fig.(??) we compare the exciton binding energies and spectra of the new implemen-  
102 tation to those of the old implementation for increasing k-grids, while keeping  $n_r$  as  
103 well as  $N_\mu^V$ ,  $N_\mu^{W_o}$ , and  $N_\mu^{W_u}$  fixed at the values above. We observe that the results  
104 converge as  $N_k$  increases. Thus, the number of interpolation points is asymptoti-  
105 cally independent of  $N_k$ . A similar behavior of ISDF was observed in (Lu & Ying, 2016).



106

107 In Fig.(??) we show the wall times for solving the BSE with the new and direct implemen-  
108 tations for increasing  $N_k$ . The new algorithm massively outperforms the direct one, and  
109 the speedup increases more than linearly with  $N_k$ . We also show the wall times for comput-  
110 ing the RPA screening with increasing  $N_k$ , which is now clearly the bottleneck in solving the BSE.



111

112 Altogether, we have implemented a new, low-scaling BSE solver and fully integrated it into  
 113 the all-electron, full-potential package exciting. We demonstrate that the new implementation  
 114 yields results equivalent to the direct solution of the BSE but with significantly reduced  
 115 computational time. Consequently, it enables more precise calculations and facilitates the  
 116 study of more complex problems.

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